

CCNA Subnetting Drill Sheet

50+ Problems with Full Worked Answers — From Baby Steps to Exam Speed

Why this exists: Subnetting shows up ALL OVER the CCNA exam, and you have to do it **fast** (about 60-90 seconds per question). Reading about it isn't enough — you have to drill it until it's automatic, like times tables. This sheet gives you 50+ problems, grouped from easy to hard, each with a **step-by-step answer**.

How to use it: Cover the answer with your hand or a piece of paper. Work the problem on scratch paper FIRST. Then check. Do **5-10 problems every single day**. In two weeks you'll be fast.

The Toolkit (Everything You Need on One Page)

The Two Magic Charts

Chart 1 — Bit values in one octet (memorize left to right):

128	64	32	16	8	4	2	1
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Chart 2 — The subnet "cheat table" (memorize this cold):

CIDR	Mask (last octet)	Block Size	Hosts (usable)
/24	0	256	254
/25	128	128	126
/26	192	64	62
/27	224	32	30
/28	240	16	14
/29	248	8	6
/30	252	4	2

Notice the pattern: **Block size doubles as you go up** (4, 8, 16, 32, 64, 128, 256), and **Mask value** = 256 – block size.

The 5 Facts You Compute for Any Subnet

For any IP + mask, you can always find these 5 things:

1. **Block size** = 256 – (the interesting octet of the mask).
2. **Network address** = the block start the IP falls into (all host bits 0).
3. **Broadcast address** = one below the next network (all host bits 1).
4. **First usable host** = network + 1.

5. **Last usable host** = broadcast – 1.

The Formulas

Number of hosts per subnet = $2^{(\text{host bits})} - 2$
Number of subnets = $2^{(\text{borrowed bits})}$
Host bits = $32 - \text{CIDR}$

The 3-Step "Where Does This IP Live?" Method

1. Find the **block size** (256 – interesting mask octet).
2. Count up in the interesting octet by the block size (0, block, 2×block...) until you pass the IP's value. The one **just below or equal** is the **network**.
3. **Broadcast** = next network – 1. Hosts sit between them.

SECTION 1 — Warm-Up: Masks, CIDR & Host Counts

(Goal: instant recall. Aim for under 15 seconds each.)

Q1. Convert /26 to a dotted-decimal mask.

Answer

255.255.255.192

/26 means 26 ones: 11111111.11111111.11111111.11000000. The last octet 11000000 = 128 + 64 = **192**.

Q2. Convert 255.255.255.240 to CIDR.

Answer

/28

240 = 11110000 = 4 ones. So 8 + 8 + 8 + 4 = **28**.

Q3. Convert /19 to a dotted-decimal mask.

Answer

255.255.224.0

/19 = 11111111.11111111.11100000.00000000. Third octet 11100000 = 128+64+32 = **224**.

Q4. How many **usable hosts** are in a /27?

Answer

30

Host bits = 32 – 27 = 5. Hosts = $2^5 - 2 = 32 - 2 =$ **30**.

Q5. How many **usable hosts** are in a /30?

Answer

2

Host bits = 32 – 30 = 2. Hosts = $2^2 - 2 = 4 - 2 =$ **2**. (This is the classic router-to-router link size.)

Q6. How many **usable hosts** are in a /22?

Answer

1022

Host bits = 32 – 22 = 10. Hosts = $2^{10} - 2 = 1024 - 2 =$ **1022**.

Q7. Convert binary 11000000 to decimal.

Answer

192

128 + 64 = **192**.

Q8. Convert decimal 224 to binary.

Answer

11100000

$128 + 64 + 32 = 224$. So the top three bits are on: **11100000**.

Q9. What is the **block size** of a **/29**?

Answer

8

/29 mask last octet = 248. Block size = $256 - 248 = 8$.

Q10. A mask ends in **.248**. What CIDR is that, and how many hosts?

Answer

/29, 6 hosts

$248 = 11111000 = 5 \text{ ones} \rightarrow 24 + 5 = /29$. Host bits = 3 $\rightarrow 2^3 - 2 = 6$.

SECTION 2 — Find the Network & Broadcast Address

(The bread-and-butter skill. Aim for under 60 seconds each.)

Q11. Given **192.168.1.77 /26**, find the network and broadcast address.

Answer

Network 192.168.1.64, Broadcast 192.168.1.127

- Block size = $256 - 192 = 64$.
- Count by 64: 0, 64, 128... 77 falls between **64 and 127**.
- Network = **.64**, next network = .128, so Broadcast = $128 - 1 = .127$.

Q12. Given **10.0.0.200 /27**, find the network and broadcast.

Answer

Network 10.0.0.192, Broadcast 10.0.0.223

- Block size = $256 - 224 = 32$.
- Count by 32: ...160, 192, 224. 200 is between **192 and 223**.
- Network = **.192**, Broadcast = $224 - 1 = .223$.

Q13. Given **172.16.5.130 /25**, find the network and broadcast.

Answer

Network 172.16.5.128, Broadcast 172.16.5.255

- Block size = $256 - 128 = 128$.
- Count by 128: 0, 128. 130 is between **128 and 255**.
- Network = **.128**, Broadcast = **.255**.

Q14. Given **192.168.10.44 /28**, find the network and broadcast.

Answer

Network 192.168.10.32, Broadcast 192.168.10.47

- Block size = $256 - 240 = 16$.
- Count by 16: 0, 16, 32, 48. 44 is between **32 and 47**.
- Network = **.32**, Broadcast = $48 - 1 = .47$.

Q15. Given **10.10.10.10 /29**, find the network and broadcast.

Answer

Network 10.10.10.8, Broadcast 10.10.10.15

- Block size = $256 - 248 = 8$.

- Count by 8: 0, 8, 16. 10 is between **8 and 15**.

- Network = **.8**, Broadcast = $16 - 1 = \mathbf{.15}$.

Q16. Given **192.168.1.100 /30**, find the network and broadcast.

Answer

Network 192.168.1.100, Broadcast 192.168.1.103

- Block size = $256 - 252 = \mathbf{4}$.
- Count by 4: ...96, 100, 104. 100 lands exactly ON **100** (a network boundary).
- Network = **.100**, Broadcast = $104 - 1 = \mathbf{.103}$.

Q17. Given **172.16.99.200 /21**, find the network and broadcast. (Now the interesting octet is the THIRD one!)

Answer

Network 172.16.96.0, Broadcast 172.16.103.255

- /21 mask = 255.255.**248**.0. The interesting octet is the **3rd**.
- Block size = $256 - 248 = \mathbf{8}$ (in the third octet).
- Count by 8 in octet 3: ...88, 96, 104. 99 is between **96 and 103**.
- Network = 172.16.**96**.0, Broadcast = 172.16.**103**.255.

Q18. Given **10.50.150.7 /20**, find the network and broadcast.

Answer

Network 10.50.144.0, Broadcast 10.50.159.255

- /20 mask = 255.255.**240**.0. Interesting octet = 3rd.
- Block size = $256 - 240 = \mathbf{16}$.
- Count by 16 in octet 3: ...128, 144, 160. 150 is between **144 and 159**.

- Network = 10.50.**144**.0, Broadcast = 10.50.**159**.255.

Q19. Given **192.168.200.55 /18**, find the network and broadcast.

Answer

Network 192.168.192.0, Broadcast 192.168.255.255

- /18 mask = 255.255.**192**.0. Interesting octet = 3rd.
- Block size = $256 - 192 = \mathbf{64}$.
- Count by 64 in octet 3: 0, 64, 128, 192, 256. 200 is between **192 and 255**.
- Network = 192.168.**192**.0, Broadcast = 192.168.**255**.255.

Q20. Given **10.1.1.1 /8**, find the network and broadcast.

Answer

Network 10.0.0.0, Broadcast 10.255.255.255

- /8 mask = 255.0.0.0. The whole 10.x.x.x block is one network.
- Network = **10.0.0.0**, Broadcast = **10.255.255.255**.

SECTION 3 — Find the Usable Host Range

(Network + 1 to Broadcast – 1. Put it all together.)

Q21. For **192.168.1.77 /26**, list the **first and last usable host**.

Answer

First 192.168.1.65, Last 192.168.1.126

Network = .64, Broadcast = .127 (from Q11). First host = $64 + 1 = \mathbf{.65}$, Last host = $127 - 1 = \mathbf{.126}$.

Q22. For 172.16.5.130 /25, list the first and last usable host.

Answer

First 172.16.5.129, Last 172.16.5.254

Network = .128, Broadcast = .255. First = **.129**, Last = **.254**.

Q23. For 10.10.10.10 /29, list the first and last usable host.

Answer

First 10.10.10.9, Last 10.10.10.14

Network = .8, Broadcast = .15. First = **.9**, Last = **.14**.

Q24. For 192.168.1.100 /30, list the first and last usable host.

Answer

First 192.168.1.101, Last 192.168.1.102

Network = .100, Broadcast = .103. First = **.101**, Last = **.102**. (Exactly 2 usable — perfect for a WAN link.)

Q25. For 172.16.99.200 /21, list the first and last usable host.

Answer

First 172.16.96.1, Last 172.16.103.254

Network = 172.16.96.0, Broadcast = 172.16.103.255. First = 172.16.96.**1**, Last = 172.16.103.**254**.

Q26. Is 192.168.1.64 a **usable host** address in the 192.168.1.0 /26 scheme? Why?

Answer

No — it's a network address.

With /26, block size 64, the subnets start at .0, .64, .128, .192. **.64** is the **network address** of the second subnet, so it can't be assigned to a device.

Q27. Is `192.168.1.191` usable in a `/26` scheme? Why?

Answer

No — it's a broadcast address.

Subnets: `.128-.191` is the third `/26` (block `.128`, next block `.192`). So `.191` is the **broadcast** of that subnet — not usable.

Q28. Two PCs: `192.168.1.70 /26` and `192.168.1.130 /26`. Can they talk **directly** (same subnet)?

Answer

No — different subnets, so they need a router.

- `.70` is in subnet **.64** (`.64-.127`).
- `.130` is in subnet **.128** (`.128-.191`). Different networks → traffic must go through a router (default gateway).

Q29. Two PCs: `10.0.0.33 /27` and `10.0.0.60 /27`. Same subnet?

Answer

No.

Block size 32 → subnets `.0-.31`, `.32-.63`, `.64-.95...`

- `.33` is in **.32** (`.32-.63`).
- `.60` is also in **.32** (`.32-.63`).

Wait — recheck: both are between 32 and 63! So they ARE in the same subnet **.32**. **Yes, same subnet — they can talk directly.**

(Lesson: always double-check by finding the block for BOTH addresses. Don't guess!)

Q30. A device is configured `192.168.5.31 /27`. What's wrong?

Answer

.31 is a broadcast address — you can't assign it to a host.

Block size 32 → subnets .0–.31, .32–.63... **.31** is the **broadcast** of the first subnet (.0–.31). Assigning it will break connectivity.

SECTION 4 — "How Many Subnets / Hosts Do I Need?" (Design)

(These are the word problems. Read carefully!)

Q31. You have **192.168.1.0 /24** and need **4 equal subnets**. What mask and how many hosts each?

Answer

Mask /26 (255.255.255.192), 62 hosts each.

- 4 subnets = 2^2 , so **borrow 2 bits**: $/24 + 2 = /26$.
- Host bits = 6 → $2^6 - 2 = 62$ hosts.
- Subnets: .0, .64, .128, .192.

Q32. You have **192.168.1.0 /24** and need **at least 50 hosts per subnet**, as many subnets as possible. What mask?

Answer

Mask /26 (255.255.255.192).

- Need 50 hosts → find host bits: $2^6 - 2 = 62 \geq 50$ ($2^5 - 2 = 30$ is too small).
- 6 host bits → mask = $32 - 6 = /26$.
- Gives 4 subnets of 62 hosts each.

Q33. You need **at least 500 hosts** in one subnet. What is the smallest mask (largest CIDR number) that works?

Answer

/23 (255.255.254.0).

- $2^9 - 2 = 510 \geq 500$ ($2^8 - 2 = 254$ too small).
- 9 host bits $\rightarrow$ mask = $32 - 9 =$ **/23**.

Q34. You need **exactly 6 subnets** from **172.16.0.0 /16**, each as large as possible. What mask?

Answer

/19 (255.255.224.0).

- 6 subnets $\rightarrow$ need $2^3 = 8$ ($2^2 = 4$ too few) $\rightarrow$ **borrow 3 bits**.
- $/16 + 3 =$ **/19**. Host bits = 13 $\rightarrow 2^{13} - 2 =$ **8190** hosts each.

Q35. How many **/30** subnets can you make from a single **/24**?

Answer

64.

From /24 to /30 you borrow 6 bits ($30 - 24$). $2^6 =$ **64** subnets. (Each /30 has 2 usable hosts — great for point-to-point links.)

Q36. How many **/26** subnets fit in a **/22**?

Answer

16.

$26 - 22 = 4$ borrowed bits $\rightarrow 2^4 =$ **16** subnets.

Q37. A company needs 200 hosts in one LAN. Will a **/25** work? What about a single **/24**?

Answer

/25 = No. /24 = Yes.

- $/25 \rightarrow 2^7 - 2 =$ **126** hosts (too small for 200).
- $/24 \rightarrow 2^8 - 2 =$ **254** hosts (fits 200).

Q38. You need a subnet for a **point-to-point link between two routers** with no wasted addresses. What mask?

Answer

/30 (or /31 for zero waste).

- **/30** gives exactly 2 usable hosts — one per router. The standard choice.
- **/31** (RFC 3021) gives 2 usable with no network/broadcast waste, used on point-to-point links by advanced setups.

Q39. From **10.0.0.0 /24**, you subnet into **/28s**. What is the **network address of the 5th subnet**?

Answer

10.0.0.64

Block size = 16. Subnets: 1st = .0, 2nd = .16, 3rd = .32, 4th = .48, **5th = .64.**

Q40. From **192.168.1.0 /24** subnetted into **/27s**, what is the **broadcast of the 3rd subnet**?

Answer

192.168.1.95

Block size = 32. Subnets: 1st .0-.31, 2nd .32-.63, **3rd .64-.95**. Broadcast of 3rd = **.95.**

SECTION 5 — VLSM (Variable Length Subnet Masks)

(Different-sized subnets from one block. The GOLDEN RULE: always assign the biggest subnet first!)

Q41. From **192.168.1.0 /24**, create subnets for: Sales (100 hosts), IT (50 hosts), HR (25 hosts), and a WAN link (2 hosts). Give the subnet + mask for each.

Answer

Assign biggest first:

Group	Hosts	Mask	Subnet	Range	Broadcast
Sales	100	/25 (126)	192.168.1.0	.1-.126	.127
IT	50	/26 (62)	192.168.1.128	.129-.190	.191
HR	25	/27 (30)	192.168.1.192	.193-.222	.223
WAN	2	/30 (2)	192.168.1.224	.225-.226	.227

Steps: Sales needs 100 → /25 (126). Start at .0, uses .0-.127. Next free = .128. IT needs 50 → /26 (62), uses .128-.191. Next free = .192. HR needs 25 → /27 (30), uses .192-.223. Next free = .224. WAN needs 2 → /30, uses .224-.227.

Q42. From **10.0.0.0 /24**, you need three subnets: A (60 hosts), B (12 hosts), C (2 hosts). Provide subnet + mask for each.

Answer

Group	Hosts	Mask	Subnet	Range	Broadcast
A	60	/26 (62)	10.0.0.0	.1-.62	.63
B	12	/28 (14)	10.0.0.64	.65-.78	.79
C	2	/30 (2)	10.0.0.80	.81-.82	.83

A: 60 → /26 (62), .0-.63. Next = .64. B: 12 → /28 (14), .64-.79. Next = .80. C: 2 → /30, .80-.83.

Q43. In Q41, what goes wrong if you assign the **WAN /30 first** at 192.168.1.0, then try to fit Sales?

Answer

You waste space and may not fit Sales cleanly (fragmentation).

If WAN takes .0-.3, Sales (/25 = 126 hosts) can't start until a 128-aligned boundary (.128). You'd push Sales to .128-.255, leaving .4-.127 chopped up and possibly unusable for the other groups. **That's why VLSM's golden rule is: biggest subnet first.**

Q44. From 172.16.0.0 /22, allocate: Building1 (500 hosts), Building2 (250 hosts), Building3 (100 hosts). Provide subnet + mask for each.

Answer

Group	Hosts	Mask	Subnet	Range	Broadcast
Bldg1	500	/23 (510)	172.16.0.0	172.16.0.1-172.16.1.255	172.16.1.255
Bldg2	250	/24 (254)	172.16.2.0	172.16.2.1-172.16.2.255	172.16.2.255
Bldg3	100	/25 (126)	172.16.3.0	172.16.3.1-172.16.3.127	172.16.3.127

Bldg1: 500 → /23 (510), uses 172.16.0.0-172.16.1.255. Next = 172.16.2.0. Bldg2: 250 → /24 (254), uses 172.16.2.0-.255. Next = 172.16.3.0. Bldg3: 100 → /25 (126), uses 172.16.3.0-.127. (All fits inside the /22!)

Q45. You subnetted 192.168.4.0 /24 with VLSM and used up to 192.168.4.223. A new team needs **20 hosts**. What subnet + mask do you give them, and does it fit?

Answer

192.168.4.224 /27, and yes it fits exactly.

- 20 hosts → /27 (30 usable). Next free address is .224.
- /27 block size 32 → uses .224-.255 (broadcast .255). Fits perfectly in the remaining space.

SECTION 6 — Rapid-Fire Mixed Drills (Exam Simulation)

(No hints. Do these on a timer — target under 60 seconds each.)

Q46. 192.168.3.194 /26 → network, broadcast, first, last?

Answer

Block 64. Subnets .0/.64/.128/.192. 194 is in **.192** (.192-.255).

- Network **192.168.3.192**, Broadcast **192.168.3.255**, First **.193**, Last **.254**.

Q47. 10.20.30.40 /28 → network, broadcast, first, last?

Answer

Block 16. 40 is between 32 and 47.

- Network **10.20.30.32**, Broadcast **10.20.30.47**, First **.33**, Last **.46**.

Q48. 172.31.100.100 /26 → network, broadcast?

Answer

Block 64. 100 is between 64 and 127.

- Network **172.31.100.64**, Broadcast **172.31.100.127**.

Q49. 192.168.15.15 /29 → network, broadcast, first, last?

Answer

Block 8. 15 is between 8 and 15.

- Network **192.168.15.8**, Broadcast **192.168.15.15**, First **.9**, Last **.14**.
- (Note: .15 is the broadcast, so the device .15 is NOT a usable host!)

Q50. 10.0.5.130 /23 → network, broadcast? (Third-octet subnetting!)

Answer

/23 mask = 255.255.**254**.0. Block size in 3rd octet = 256 – 254 = **2**. Count by 2 in octet 3: 0, 2, 4, 6... 5 is between **4 and 5**.

- Network **10.0.4.0**, Broadcast **10.0.5.255**.

Q51. 192.168.1.222 /27 → is it a host, network, or broadcast?

Answer

Usable host.

Block 32. Subnet containing 222 is **.192** (.192-.223). Network = .192, Broadcast = .223. Since 222 is between .193 and .222, it's a **usable host**.

Q52. 172.16.14.0 /23 → how many usable hosts, and what's the broadcast?

Answer

510 hosts; broadcast 172.16.15.255.

/23 → host bits 9 → $2^9 - 2 = 510$. Block size 2 in 3rd octet → 14 and 15 belong together (14-15), so range is 172.16.14.0-172.16.15.255, broadcast **172.16.15.255**.

Q53. 10.10.10.100 /25 → which subnet (network address)?

Answer

10.10.10.0

Block 128. Subnets: .0 (.0-.127) and .128 (.128-.255). 100 is in **.0**. Network = **10.10.10.0**.

Q54. 192.168.100.100 /30 → network, broadcast, both usable hosts?

Answer

Block 4. Count by 4: ...96, 100, 104. 100 is a boundary.

- Network **192.168.100.100**, Broadcast **192.168.100.103**, Hosts **.101 and .102**.

Q55. You see 255.255.255.252. How many of these subnets fit in a /24, and what are they good for?

Answer

64 subnets; good for point-to-point router links.

255.255.255.252 = **/30**. From /24 to /30 you borrow 6 bits → $2^6 = 64$ subnets, each with **2 usable hosts**. Perfect for router-to-router (WAN) links where you only need one address on each end.

SECTION 7 — Boss Level: Full Scenario Problems

(These mimic the harder simulation-style exam questions. Take your time.)

Q56. A router interface is configured with `192.168.1.65 /26`. A host on that LAN is set to `192.168.1.130 /26` with gateway `192.168.1.65`. The host can't reach the internet. What's the problem and the fix?

Answer

Problem: the host and its gateway are in different subnets.

- Gateway `.65 /26` is in subnet **.64** (`.64-.127`).
- Host `.130 /26` is in subnet **.128** (`.128-.191`).

The host can't reach a gateway that isn't on its own subnet, so it can't route out. **Fix:** give the host an address in `.64-.127` (e.g., `192.168.1.66`), OR move the gateway into the host's subnet. They must share the same subnet.

Q57. You're given `192.168.50.0 /24` and must support **6 subnets**, each needing **25 usable hosts**. Give the mask and list all 6 network addresses.

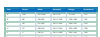
Answer

Mask /27 (255.255.255.224).

- 25 hosts → /27 gives 30 usable .
- /27 also gives $2^3 = 8$ subnets (≥ 6 needed) .
- Block size 32 → networks: **.0, .32, .64, .96, .128, .160** (first 6), plus .192, .224 spare.

Q58. `10.1.1.0 /24` needs these, biggest-first (VLSM): Subnet A 120 hosts, B 60 hosts, C 30 hosts, D 12 hosts, plus two /30 WAN links (E, F). List each subnet + mask + range.

Answer



Everything fits inside the /24 with room to spare (.248-.255 free).

Q59. Summarize (aggregate) these four networks into ONE route: 172.16.0.0/24, 172.16.1.0/24, 172.16.2.0/24, 172.16.3.0/24.

Answer

172.16.0.0 /22

The four /24s (.0, .1, .2, .3 in the 3rd octet) share the top 22 bits. In binary the 3rd octet: 00000000, 00000001, 00000010, 00000011 — the first **6 bits** are identical, and $16 + 6 = 22$. So the summary is **172.16.0.0/22** (covers .0 through .3). This is **route summarization**, which shrinks routing tables.

Q60. Summarize 192.168.8.0/24 through 192.168.15.0/24 into one route.

Answer

192.168.8.0 /21

Third octet values 8-15 in binary: 00001000 ... 00001111. The first **5 bits** (00001) are common → $16 + 5 = 21$. Summary = **192.168.8.0/21** (covers .8 through .15).

The One-Page Speed Method (Print This!)

When you see any "IP + CIDR" question, do this in your head, every time:

```
STEP 1: Which octet is "interesting"?
        /1-8 → 1st octet
        /9-16 → 2nd octet
        /17-24 → 3rd octet
        /25-32 → 4th octet

STEP 2: Block size = 256 - (mask value in that octet)

STEP 3: Count 0, block, 2xblock... until you pass the IP.
        The one at/just-below = NETWORK.

STEP 4: Broadcast = next network - 1
        First host = network + 1
        Last host = broadcast - 1

STEP 5: Usable hosts = 2^(32-CIDR) - 2
```

Final Tips

1. **Memorize the cheat table** (/24-/30 with block sizes and host counts). It saves you 30 seconds every question.
2. **Always find the block for BOTH addresses** when asked "same subnet?" — don't eyeball it.
3. **Watch out for network & broadcast addresses** disguised as host answers (a favorite trick).
4. **VLSM golden rule:** biggest subnet first, every time.
5. **Do 10 problems a day.** Speed comes only from reps. In two weeks this will feel easy.

You've got this. Now go drill!

— End of Subnetting Drill Sheet —